



PROFESSIONAL SUMMARY

Detail-oriented **Software Development Engineer in Test (SDET)** with hands-on experience in **backend automation, distributed system validation, and cloud-native test infrastructure**. Skilled in **Python, PyTest, REST API testing, CI/CD pipelines, and Linux/Unix systems**. Demonstrated success in designing and executing **manual and automated test cases**, improving performance and reliability across **large-scale enterprise and distributed environments**. Strong collaborator with Agile teams, passionate about solving **system-level challenges** and delivering **high-quality scalable software**.

WORK EXPERIENCE

Apprentice Engineer – Palo Alto Networks (2023 – Present)

- Designed and executed **manual and automated test cases** for **PAN-OS features (L2/L3 networking, security policies)** ensuring functional and system-level coverage.
- Developed **PyTest-based automation frameworks** and Python scripts for configuration and regression testing across distributed firewall systems.
- Performed **REST API testing** and **CLI automation** using Python (Paramiko) to validate customer use-cases.
- Worked with **AWS, Docker, Jenkins, and Kubernetes** for CI/CD deployment and validation in distributed environments.
- Conducted **integration, performance, and system testing**, validating end-to-end scenarios over TCP/IP, HTTP, and DNS protocols.
- Collaborated with cross-functional **Dev, QA, and Support teams** in an **Agile/Scrum** environment using **Jira** for defect tracking and reporting.

TECHNICAL SKILLS

- Programming & Automation:** Python, Java, JavaScript, PyTest, Paramiko (PARTS), REST API Testing, Postman, Django APIs
- Testing & QA:** Test Plans, Test Case Design, Test Execution, Functional Testing (Integration/System), API Testing, Performance Testing, Manual Testing
- Networking & Protocols:** TCP/IP, HTTP, DNS, SSL, Client-Server Communication, Load Testing
- Cloud & DevOps:** AWS (EC2, S3, Lambda), Docker, Jenkins, Kubernetes (basic), Git, GitHub, CI/CD Pipelines
- Databases & Caching:** MySQL, MongoDB, Redis (basic), SQL Optimization
- Systems & Tools:** Linux/Unix, Bash Scripting, Jira, Confluence, Kafka (basic)

PROJECTS

TechSupport File Analyzêr (TSF Analyzer)

Python | Django | MySQL | Automation

- Built a Django-based SaaS web application to automate analysis of configuration logs and scale metrics from customer PAN-OS devices.
- Implemented REST APIs, automated workflows, and MySQL integration for high-volume data handling in distributed test setups.
- Designed functional and performance test cases replicating real-world network scenarios using TCP/IP and HTTP protocols, reducing manual analysis effort by 70%.

Panorama Characterization Tool

Python | Paramiko | Matplotlib | Jira API

- Automated CLI and API testing using a Paramiko-based Python framework (PARTS) to validate Panorama features.
- Created test cases for distributed system components, integrated with Jira APIs for automated bug tracking and reporting.
- Generated visual reports using Matplotlib to analyze system behavior across builds and releases.

Processes Leak Detection System

Python | Pandas | NumPy

- Developed an automation system to identify memory leaks, file descriptor leaks, and CPU anomalies from >50,000 process reports.
- Applied Pandas/NumPy for anomaly detection and trend analysis, improving test efficiency and scalability validation for distributed workloads.

ACCOMPLISHMENTS

- Ranked **top-10 innovation** out of 150 projects; showcased TSF Analyzer at company-wide All Hands
- Recognized for test automation efficiency improvements and framework contributions.